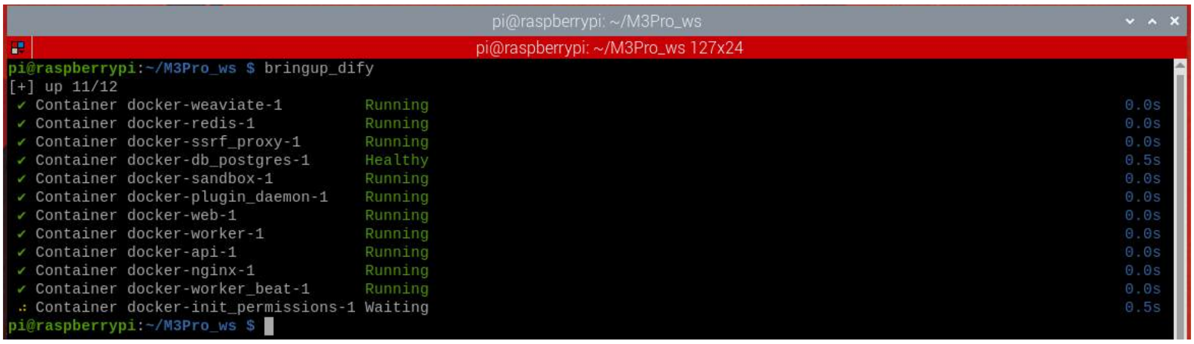
Core Module Testing Tools

1. Course Content

- multi_brains provides minimal test programs for testing the core modules of a large model, used to quickly locate problems in abnormal situations.

2. Start the Dify service



```
pi@raspberrypi: ~/M3Pro_ws
pi@raspberrypi: ~/M3Pro_ws 127x24
pi@raspberrypi:~/M3Pro_ws $ bringup_dify
[+] up 11/12
 ✔ Container docker-weaviate-1          Running     0.0s
 ✔ Container docker-redis-1             Running     0.0s
 ✔ Container docker-ssrf_proxy-1        Running     0.0s
 ✔ Container docker-db_postgres-1       Healthy     0.5s
 ✔ Container docker-sandbox-1           Running     0.0s
 ✔ Container docker-plugin_daemon-1     Running     0.0s
 ✔ Container docker-web-1               Running     0.0s
 ✔ Container docker-worker-1            Running     0.0s
 ✔ Container docker-api-1               Running     0.0s
 ✔ Container docker-nginx-1             Running     0.0s
 ✔ Container docker-worker_beat-1       Running     0.0s
 ∴ Container docker-init_permissions-1 Waiting      0.5s
pi@raspberrypi:~/M3Pro_ws $
```

3. Navigate to the test tool path

For Raspberry Pi 5 users, if the ROS container is not running, you need to start it first.

```
bringup_m3
```

Open a terminal inside the container.

```
exec_m3
```

```
cd ~/yahboomM3_ws/code_ws/src/multi_brains/test
ls
```

- The test tools are as follows:

```
test_PiperTTS.py         test_TongyiASR.py  test_copyright.py        test_flake8.py   test_xunfeiASR.py
test_SenseVoiceSmall.py  test_TongyiTTS.py  test_dify_connection.py  test_pep257.py   test_xunfeiTTS.py
root@raspberrypi:~/M3Pro_ws/src/multi_brains/test#
```

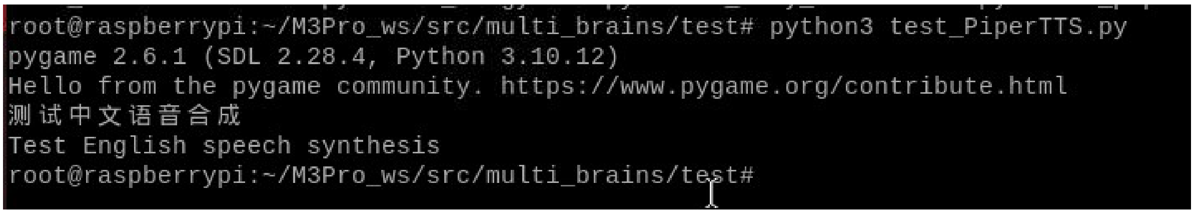
4. Test Local Speech Synthesis Functionality

Only ORIN Mainboard

- Test local speech synthesis functionality for Chinese and English in sequence.

```
python3 test_PiperTTS.py
```

After running, the speech synthesis results for Chinese and English will be played sequentially.



```
root@raspberrypi:~/M3Pro_ws/src/multi_brains/test# python3 test_PiperTTS.py
pygame 2.6.1 (SDL 2.28.4, Python 3.10.12)
Hello from the pygame community. https://www.pygame.org/contribute.html
测试中文语音合成
Test English speech synthesis
root@raspberrypi:~/M3Pro_ws/src/multi_brains/test#
```

5. Test Local Speech Recognition Functionality

Only ORIN Mainboard

- After running, test preset Chinese and English audio recordings sequentially.

```
python3 test_SenseVoiceSmall.py
```

After running, the speech recognition results for the test audio will be printed.

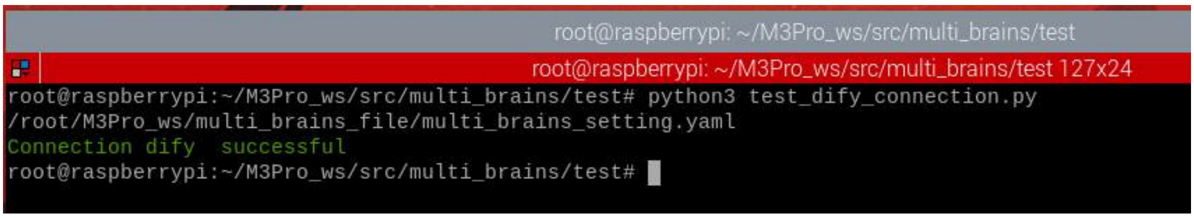
```
root@raspberrypi: ~/M3Pro_ws/src/multi_brains/test
root@raspberrypi: ~/M3Pro_ws/src/multi_brains/test 127x24
root@raspberrypi:~/M3Pro_ws/src/multi_brains/test# python3 test_SenseVoiceSmall.py
=========================================测试本地中文语音识别===============================================
funasr version: 1.2.6.
SenseVoiceSmall ASR_Engine initialized successfully
rtf_avg: 0.463: 100%|██████████| 1/1 [00:01<00:00,  1.36s/it]
你好我是巨身智能机器人
=============================Test local English speech recognition===========================
funasr version: 1.2.6.
SenseVoiceSmall ASR_Engine initialized successfully
rtf_avg: 0.353: 100%|██████████| 1/1 [00:01<00:00,  1.21s/it]
hello i'm embodied intelligent robot
root@raspberrypi:~/M3Pro_ws/src/multi_brains/test#
```

6. Testing Robot Access to Dify

- When the program displays "Model service unavailable," and it's unclear whether the robot's inability to access Dify is due to an incorrect network address, you can test:

```
python3 test_dify_connection.py
```

If the robot displays the following information normally, it proves that the robot's connection to Dify is normal. Otherwise, it may be that Dify is not started or the Dify program has crashed and needs to be restarted.



```
root@raspberrypi: ~/M3Pro_ws/src/multi_brains/test
root@raspberrypi: ~/M3Pro_ws/src/multi_brains/test 127x24
root@raspberrypi:~/M3Pro_ws/src/multi_brains/test# python3 test_dify_connection.py
/root/M3Pro_ws/multi_brains_file/multi_brains_setting.yaml
Connection dify  successful
root@raspberrypi:~/M3Pro_ws/src/multi_brains/test#
```

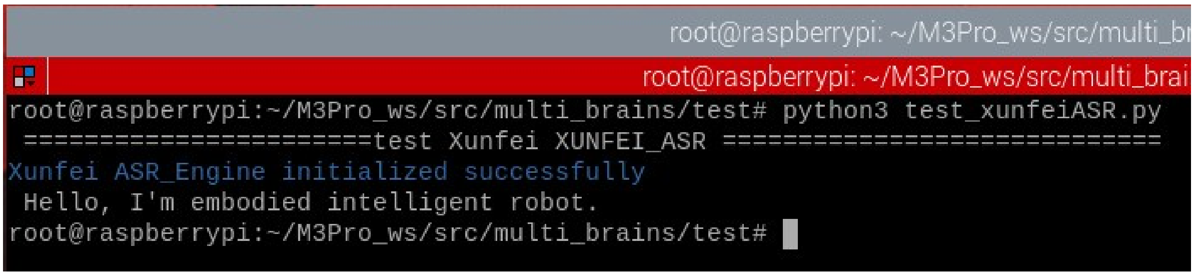
7. Online Voice Service for International Users

7.1 iFlytek Speech Recognition Service

- If online speech recognition encounters errors, you can test iFlytek's speech recognition service separately.

```
python3 test_xunfeiASR.py
```

- After running, it will perform speech recognition tests on the preset test audio.



7.2 iFlytek Speech Synthesis Service

- If online speech synthesis encounters errors, you can test iFlytek's speech recognition service separately.

```
python3 test_xunfeiTTS.py
```

After running, it will first perform speech synthesis on the default text, and then play the corresponding audio.